Input Form (IF)

Score: 4/5 — Minor gaps | Confidence: medium

Benchmark: *MASSIVE* | ***Context:*** *Ethiopian Intercity Travelers — Amharic-Language Booking Bot*

Input Form definition: Whether the input signal encoding (text, audio, image parameters) matches deployment conditions.

Each section below is followed by an evidence table. Three types of evidence are cited: **[QN]** — verbatim quotes extracted from the benchmark paper; **[WEB-N]** — web sources consulted during assessment; **[DN]** / **[DATASET-DN]** — sampled datapoints from the benchmark dataset, with an interpretation note.

Justification

Modality alignment is strong: MASSIVE is text-only [Q3] and Amharic is included as one of 51 languages, with 21 distinct scripts represented [Q25] including Ethiopic/Ge'ez fidel as a 'unique' script [Q27]. The deployment is also text-only smartphone input in Ethiopic script. Script diversity was explicitly sought 'to drive experimentation in tokenization and normalization' [Q20]. The paper acknowledges 'our collection system did not have a robust method to preserving or denoting native tokenization practices' [Q109], leaving slot-boundary handling for Amharic's morphologically rich, fusional structure as an open question (gap_id 8 PARTIAL — amseg and HornMorpho exist but not evaluated against MASSIVE Amharic). Amharic uses spaces as word delimiters, so unlike Japanese/Chinese/Khmer it does not require character-level splitting [Q80, Q92, Q96]. The IF dimension is rated LOWER priority and modality match is good, but documentation of Amharic-specific tokenization quality is absent.

ID	Evidence
Q3	“With the English seed data included, there are 587k train utterances, 104k dev utterances, 152k test utterances, and 153k utterances currently held out for the MMNLU-22 competition, which will be released after the competition.” (p.1)
Q20	“Fifth, we examined the script of each language, seeking to increase script diversity to drive experimentation in tokenization and normalization.” (p.3)
Q25	“There are 21 distinct scripts used in the dataset.” (p.3)
Q27	“The Arabic script is used for three languages, the Cyrillic script for two languages, and the remaining 18 languages have "unique" scripts, in the sense that only one language in the dataset uses that script.” (p.3)
Q80	“Three of these written languages, Japanese, Chinese (Traditional), and Chinese (Simplified), do not use spaces anywhere except to identify the end of a sentence. For these languages, we separate each character in the unlabeled input with a whitespace.” (p.7)
Q92	“For Khmer, the workers had a difficult time adapting their translations and localizations to properly-slotted outputs given the space-optional nature of the language.” (p.8)
Q96	“character spacing was necessary in order to properly assign the slots to the right characters.” (p.8)
Q109	“Fifth, our collection system did not have a robust method to preserving or denoting native tokenization practices—some languages do not separate with whitespace, while others do but there is no set practice.” (p.9)

Strengths

- Text-only modality matches deployment exactly [Q3]
- Amharic Ethiopic script is included as a 'unique' script category [Q27]; script diversity was an explicit design goal [Q20]
- Amharic uses space-delimited words, sidestepping the character-spacing issues that affected Japanese/Chinese/Khmer [Q80, Q92] — the 'default spacing provided by annotators' [Q82] applies

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Q20	“Fifth, we examined the script of each language, seeking to increase script diversity to drive experimentation in tokenization and normalization.” (p.3)
Q27	“The Arabic script is used for three languages, the Cyrillic script for two languages, and the remaining 18 languages have "unique" scripts, in the sense that only one language in the dataset uses that script.” (p.3)

Q80	"Three of these written languages, Japanese, Chinese (Traditional), and Chinese (Simplified), do not use spaces anywhere except to identify the end of a sentence. For these languages, we separate each character in the unlabeled input with a whitespace." (p.7)
Q82	"We use the default spacing provided by annotators for all other languages." (p.7)
Q92	"For Khmer, the workers had a difficult time adapting their translations and localizations to properly-slotted outputs given the space-optional nature of the language." (p.8)

Checklist

Checklist item definitions:

ID	Assessment Question
IF-1	Compare signal distributions between source and target
IF-2	Check if regional infrastructure supports data capture specs
IF-3	Identify domain-specific form differences
IF-4	Document form mismatches harming external validity

Checklist responses:

IF-1: Both source and target are text in Ethiopic script; no signal-distribution mismatch on modality. Smartphone keyboard input methods (fidel, phonetic IME, occasional Latin-script Amharic) are not specifically validated by MASSIVE — INSUFFICIENT DOCUMENTATION on input-method robustness.

IF-2: Ethiopian smartphone infrastructure supports Ethiopic input; mobile penetration reached 61.4% Q1 2024 [WEB-9]; smartphone penetration ~15% nationally [WEB-8]. The deployment population is a self-selected app-user cohort with sufficient infrastructure.

IF-3: Amharic morphological richness (cliticization, root-and-pattern morphology — regional context) means slot boundaries may span morphologically complex tokens; MASSIVE acknowledges 'collection system did not have a robust method to preserving or denoting native tokenization practices' [Q109], leaving slot-boundary quality for Amharic unverified. amseg and HornMorpho tokenizers exist [WEB-17, WEB-19] but were not used.

IF-4: Form mismatches are minor: text-modality alignment is strong, but Amharic-specific tokenization quality and IME variants (fidel vs. romanized input) are not documented; this is a moderate but not severe external-validity risk.

ID	Evidence
Q1	"MASSIVE contains 1M realistic, parallel, labeled virtual assistant utterances spanning 51 languages, 18 domains, 60 intents, and 55 slots." (p.1)
Q3	"With the English seed data included, there are 587k train utterances, 104k dev utterances, 152k test utterances, and 153k utterances currently held out for the MMNLU-22 competition, which will be released after the competition." (p.1)
Q82	"We use the default spacing provided by annotators for all other languages." (p.7)
Q109	"Fifth, our collection system did not have a robust method to preserving or denoting native tokenization practices—some languages do not separate with whitespace, while others do but there is no set practice." (p.9)
WEB-8	Smartphone penetration ~15% nationally — deployment cohort is self-selected high-digital-literacy users (https://birrmetrics.com/ethiopia-narrows-mobile-gender-gap-to-24-but-smartphone-access-for-women-remains-just-6/)
WEB-9	Mobile penetration 61.4% Q1 2024; 4G expanded to 45+ cities — adequate infrastructure for text-modality deployment (https://www.connectingafrica.com/connectivity/ethiopia-s-telecoms-liberalization-makes-progress-omdia)
WEB-17	amseg Amharic segmenter/tokenizer (Yimam et al. 2021) exists (https://github.com/EthioNLP/Ethiopian-Language-Survey)
WEB-19	HornMorpho rule-based morphological analyzer used in Ethiopian NLP pipelines (https://ethionlp.github.io/)

Information Gaps

- Whether MASSIVE Amharic utterances were tokenized with amseg or default whitespace
- Per-language slot-F1 sensitivity to tokenization choice for Amharic specifically

Requires Expert Verification

- Whether smartphone IME variation (fidel vs. phonetic vs. Latin) introduces real-world input distributions not represented in MASSIVE

Remediation

Gap 1: Amharic tokenization for slot boundaries not robustly handled by MASSIVE [Q109]; morphological richness may obscure slot spans

Recommendation: Re-tokenize Amharic training/eval data using amseg [WEB-17] or HornMorpho [WEB-19] morphological segmentation, and compare slot-F1 against MASSIVE's default whitespace tokenization to quantify the tokenization-quality effect on slot boundary detection.

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Q109	“Fifth, our collection system did not have a robust method to preserving or denoting native tokenization practices—some languages do not separate with whitespace, while others do but there is no set practice.” (p.9)
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